

Mathew Schwartzman

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PROFESSIONAL EXPERIENCE

Sea Machines Robotics, Boston, MA

April 2024 – March 2026

Full-Stack Autonomy Engineer

- Worked with MIT Marine Autonomy Lab to improve MOOS-IvP middleware
- Architected upgrades to loiter, and collaborative swarm behaviors
- Led software validation with test vessel in Boston Harbor/Quincy Bay
- Built new multi-agent (collaborative and adversarial) behaviors based in MOOS-IvP

NEREID Ocean Conservation, Weston, MA

February 2025 - Present

Senior Scientist and Cofounder

- Built and presented maritime simulation framework for faster-than-realtime robotic testing
- Designed and presented low-cost CTD and web platform for hosting citizen data
- Spearheaded ocean research program using ArduPilot-based marine vehicles and custom sensors

Woods Hole Oceanographic Institution, Falmouth, MA

December 2022 – October 2024

Robotics Software Engineer II

- Supported NSF-funded deep-submergence operations in at-sea research on R/V Thomas G Thompson and NOAA's Nancy Foster
- Lead project to improve Navest software for heterogeneous vehicle ops on EV Nautilus
- Built INS autonomy ROS node to autonomously tune EKF
- Enhanced ship-to-shore data pipeline for oceanographers using popular webtool SeaLog
- Wrote linux-based OS and device tree overlay for the OOI 3D Acoustic Telescope

6 River Systems, Waltham, MA

October 2018 – July 2022

Robotics System Integration Engineer

- Lead TechOps integration efforts in customer-facing role
- Developed software for high-level multi-agent management
- Architected custom app for ML-model training/deployment
- Developed Unity VR software to allow company to meet during COVID-19

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

February 2025 - May 2025

Advanced Study Program

Unmanned Marine Vehicle Autonomy, Sensing and Communications

Worcester Polytechnic Institute, Worcester, MA

August 2013 - May 2018

Bachelor of Science, Robotics Engineering

Unified Robotics, Calculus I-IV, Linear Algebra, Differential Equations, Electrical Engineering, Artificial Intelligence, Dynamics, Controls

PUBLICATIONS

A modular interface for multi-agent Marine Autonomy Simulation

October 2025

Schwartzman, Mathew, Charlie Benjamin. "A modular interface for multi-agent Marine Autonomy Simulation." In OCEANS 2025 Great Lakes, pp. 1-4. IEEE, (2025).

Toward Crowd-Sourced Technologies for Global Citizen-Science Water Analysis

June 2025

Schwartzman, Mathew, W. Dziuban, C. Christelis, A. Markoski and E. Ling, "Toward Crowd-Sourced Technologies for Global Citizen-Science Water Analysis," OCEANS 2025 Brest, BREST, France, 2025, pp. 1-4, IEEE, (2025)

Toward standard interfaces for high-level Autonomy Simulation with MOOS-IvP

May 2025

Schwartzman, Mathew. " Toward standard interfaces for high-level Autonomy Simulation with MOOS-IvP." In ICRA 2025 Atlanta. IEEE, (2025).

SKILLS

Programming languages: C, C++, QT5, Unity, MAVLink, Javascript, Python, ROS/ROS2, MOOS-IvP, MatLab, Tensorflow, ReactJS, MongoDB, PostgreSQL, Docker, Java

Computer software/ frameworks: Git, Jira/Trello Task Management, Linux OS, Windows OS, Mac OSX, Arduino, ArduPilot, Solidworks, Autodesk Fusion 360, VS Code, LucidChart, Microsoft Office